L27: The chi-squared goodness of fit

Goals for this lecture

- Goodness of fit: looking at one variable with multiple categories
- Introduce the chi-squared

One variable with multiple categories

One categorical variable with more than 2 categories

- With one sample of a categorical variable with two categories (binary, yes/no) we tested that the proportion was equal to a hypothesized null (one sample test of proportions)
- When we had two samples or two groups we compared the difference in proportions.
- What do we do with one sample and a categorical variable when there are more than 2 categories?

One categorical variable with more than 2 categories

The general recipe we will follow for these types of variables is:

- estimate how many observations we would expect in each category under our null hypothesis
- compare the number of observations in each category to the expected value
- summarize these differences and compare them to a theoretical distribution

Jury Selection example

Suppose that the following number of people were selected for jury duty in the previous year, in a county where jury selection was supposed to be random.

Ethnicity	White	Black	Latinx	Asian	Other	Total
Number selected	1920	347	19	84	130	2500

You read online about concerns that jury was not selected randomly. How can you test this evidence?

- Example derived from this video.

Jury Selection example

Consider the distribution of race/ethnicity in the county overall:

Ethnicity	White	Black	Latinx	Asian	Other	Total
% in the population	42.2%	10.3%	25.1%	17.1%	5.3%	100%

How do we determine the counts that are expected (E) under the assumption that selection was random?

Ethnicity	White	Black	Latinx	Asian	Other	Total
Expected count						2500

Jury Selection example

Ethnicity	White	Black	Latinx	Asian	Other	Total
% in the population	42.2%	10.3%	25.1%	17.1%	5.3%	100%

- To fill in the table, multiply the total size of the jury by the % of the population of each race/ethnicity:

Expected counts under the assumption that selection is random from the county:

Ethnicity	White	Black	Latinx	Asian	Other	Total
Expected	2500 $\times$	2500 $\times$	2500 $\times$	2500 $\times$	2500 $\times$	2500
count	0.422	0.103	0.251	0.171	0.053	
$=$	1055	257.5	627.5	427.5	132.5	2500

Jury Selection example

How far off does the observed counts of race/ethnicities in the sample differ from what we would expect if the jury had been selected randomly?

Jury Selection example

Here are the counts we observed (O):

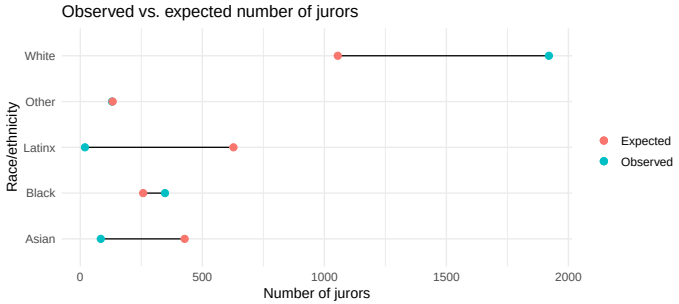
Ethnicity	White	Black	Latinx	Asian	Other	Total
Observed count	1920	347	19	84	130	2500

Which we can compare to our expected (E):

Ethnicity	White	Black	Latinx	Asian	Other	Total
Expected count	1055	257.5	627.5	427.5	132.5	2500

Jury Selection example

What is the chance of deviations of these magnitudes (or larger) under the null hypothesis?



Jury Selection example

- Recall the usual form of the test statistic:

$$\frac{estimate - null}{SE}$$

- We want an estimate that somehow quantifies how different the observed counts (O) are from the expected counts (E) across the 5 race/ethnicities.

The Chi-square test statistic

The chi^2 test statistic quantifies the magnitude of the difference between observed and expected counts under the null hypothesis. It looks like this:

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

- k is the number of cells in the table. Here, k is the number of race/ethnicity groups. That is, $k = 5$
- O_i is the observed count for the ith group (here race/ethnicity)
- E_i is the expected count for the ith group
- χ^2 is a distribution, like t or Normal.

The Chi-square test statistic

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

- The numerator measures the squared deviations between the observed (O) and expected (E) values. Bigger deviations will make the test statistic larger (which means that its corresponding p-value will be smaller)
- The denominator makes this magnitude *relative* to what we expect. This adjusts for the different magnitude of expected counts. For example, with our example, we would *expect* the number of white jurors to be close to 1055, but we would expect the number of Latinx jurors to be close to 628. Therefore, we divide by these expectations, so that a shortfall of 100 Latinx jurors counts for more than a shortfall of 100 white jurors.

Conditions to perform a chi-square test

- Fixed n of observations
- All observations are independent of one another. What does this mean in our jury example?
- Each observation falls into just one of the k mutually exclusive categories
- The probability of a given outcome is the same for each observation.

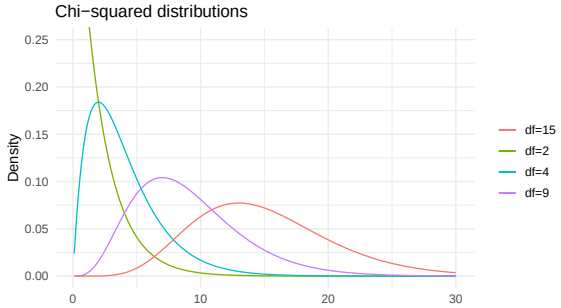
Counts requirement

- At least 80% of the cells have 5 or more expected observations ($E_i \geq 5$ for $\geq 80\%$ of the cells)
- All k cells have expected counts > 1 ($E_i > 1$)

The Chi-Square distribution

The Chi-square distribution

The chi-square distribution is a new distribution to us. Like the t-distribution, the chi-square distribution has a degrees of freedom parameter that changes the shape of the distribution. The degrees of freedom is equal to the number of groups (here, race/ethnicities) - 1. Or, $df = k - 1$.



The shape of the Chi-square

- As the `df` is increased, the distribution's central tendency moves to the right.
- This means that there will be more probability out in the right tail when the degrees of freedom is higher.
- The chi-square distribution is also positive. We only ever compute upper tail probabilities for the chi-square test because there is only one form to the H_a.

Back to the jury example

State the null and alternative hypotheses.

- The null hypothesis is that the proportions of each race/ethnicity in the jury pool is the same as the proportion of each group in the county. That is:

$H_0 : p_{white} = 42.2\%,\ p_{black} = 10.3\%,\ p_{latinx} = 25.1\%,\ p_{asian} = 17.1\%,\ p_{other} = 5.3\%$

H_a : At least one of p_k is different than specified in H_0, for k being one of white, black, latinx, asian, or other.

Back to the jury example

Calculate the chi-square statistic using the jury data.

Ethnicity	White	Black	Latinx	Asian	Other	Total
O	1920	347	19	84	130	2500
E	1055	257.5	627.5	427.5	132.5	2500

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

$$\chi^2 = \frac{(1920-1055)^2}{1055} + \frac{(347-257.5)^2}{257.5} + \frac{(19-627.5)^2}{627.5} + \frac{(84-427.5)^2}{427.5} + \frac{(130-132.5)^2}{132.5}$$

$$\chi^2 = 709.218 + 31.10777 + 590.0753 + 276.0053 + 0.04716981$$

$$\chi^2 = 1606.454$$

Back to the jury example

Calculate the p-value (what is the appropriate degrees of freedom?).

```
pchisq(q = 1606.454, df = 4, lower.tail = FALSE)
```

```
## [1] 0
```

The probability of seeing this pool of people chosen for jury duty under the null hypothesis of random sampling from the county is so small that R rounded the p-value to 0!

Chi-square test in R

Run the chi-square test using the `chisq.test` command in R.

```
# x = observed counts, p = probabilities under the null
chisq.test(x = c(1920, 347, 19, 84, 130),
           p = c(.422, .103, .251, .171, .053))
```

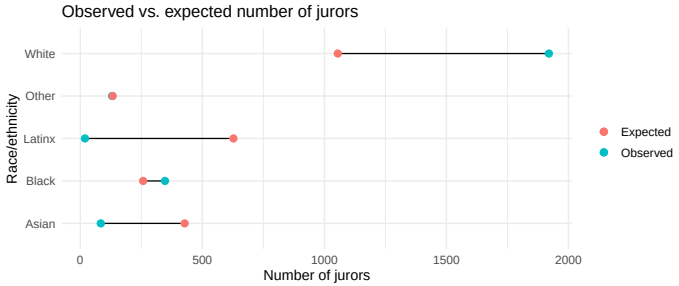
```
##
##  Chi-squared test for given probabilities
##
## data:  c(1920, 347, 19, 84, 130)
## X-squared = 1606.5, df = 4, p-value < 2.2e-16
```

Interpretation

- Which race/ethnicities appear to deviate the most from what was expected under the null hypothesis?
  - Compare the proportion observed vs. proportion expected
  - Compare the count observed vs. the count expected
  - Compare the 5 contributions to the chi-square test from each race/ethnicity. We see that whites, Latinx, and Asians contribute the most to the χ^2 statistic. This agrees with what we saw in the data visualization in terms of the size of the gaps between observed and expected counts.

The picture behind the test statistic

The biggest gaps below are exactly the categories that contribute most to χ^2.



Statistics is Everywhere

COVID vaccine roll out

U.S. South Is Fast Closing Country's Widest Black Vaccine Gaps: Covid-19 Tracker

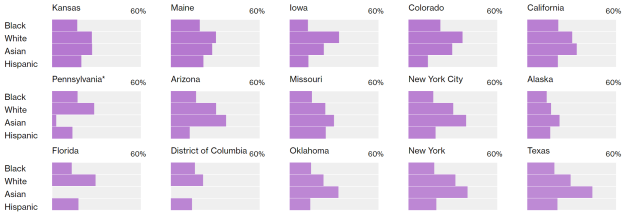
Bloomberg's tracker now shows which states are making the most week-by-week progress in closing their racial vaccination gaps.

Updated: April 16, 2021, 3:00 AM PDT

COVID vaccine roll out

One hypothesis in the article is that "Now that almost all adults in the U.S. are eligible for Covid vaccines, the racial and ethnic disparities among those getting shots should narrow."

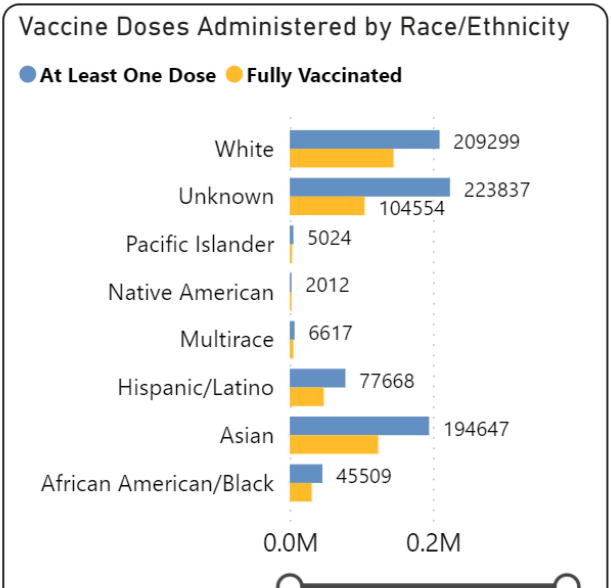
Visualization by race/ethnicity



How do we know if the variability is due to chance?

How could we see what the distribution is like for Alameda County?

From Alameda County Data website 16 April 2021



Example 2: Alameda

Race/ethnicity	Doses given	% of population
White	209,299	30.6%
Pacific Islander	5,024	0.9%
Native American	2,012	1.1%
Multirace	6,617	5.4%
Latinx	77,668	18.7%
Asian	194,647	32.3%
AA/Black	45,509	11.0%
Total	540,776	100%

We remove "unknown" for now and subtract them from the total, leaving 540,776 individuals vaccinated with known race/ethnicity.

Example 2: Alameda

Now compare the share of the population to the share of the doses:

Race/ethnicity	% of population	% of doses
White	30.6%	38.7%
Pacific Islander	0.9%	0.9%
Native American	1.1%	0.4%
Multirace	5.4%	1.2%
Latinx	18.7%	14.4%
Asian	32.3%	36.0%
AA/Black	11.0%	8.4%

Example 2: Alameda Chi-square test in R

Run the chi-square test using the `chisq.test` command in R.

```
doses <- c(209299, 5024, 2012, 6617, 77668, 194647, 45509)
pop_p <- c(.306, .009, .011, .054, .187, .323, .110)
chisq.test(x = doses, p = pop_p)
```

```
##
##  Chi-squared test for given probabilities
##
## data:  doses
## X-squared = 42692, df = 6, p-value < 2.2e-16
```

Example 3: cheating at dice?

Suppose there is a game in which the objective is to roll as many sixes as possible using 3 dice. Over 100 rolls, one of the players seems to be winning quite often, and we see the following.

Number of 6s	0	1	2	3
Observed rolls	47	35	15	3

We suspect they are using a loaded die or cheating in some way.

Are they cheating? Or just lucky (within the bounds of chance)?

Example derived from this site

Example 3: cheating at dice?

What would we expect?
The rolls of dice should follow a binomial distribution (# of successes in # trials)

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

What is p here? What is n?

Example 3: cheating at dice?

Remember `dbinom`?

`dbinom(x, size, prob)`

This function calculates the probability of observing `x` successes when $X \sim Binom(n, p)$

```
Expected <- dbinom(0:3, size = 3, prob = 1/6)
Expected
```

```
## [1] 0.57870370 0.34722222 0.06944444 0.00462963
```

Example 3: cheating at dice?

Number of 6s	0	1	2	3
Observed rolls	47	35	15	3
Expected rolls	57.9	34.7	6.9	0.46

Keep an eye on that last expected count.

Example 3: cheating at dice?

```
chisq.test(x = c(47, 35, 15, 3), p = Expected)
```

```
## Warning in chisq.test(x = c(47, 35, 15, 3), p = Expected):
## Chi-squared approximation may be incorrect
```

```
##
##  Chi-squared test for given probabilities
##
## data:  c(47, 35, 15, 3)
## X-squared = 25.292, df = 3, p-value = 1.342e-05
```

Example 3: check the conditions!

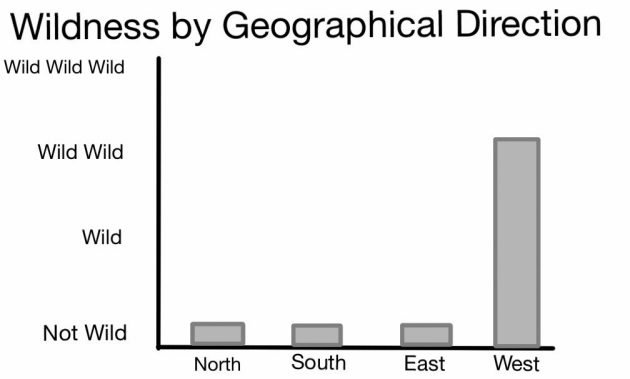
- R warned us. Look back at the expected counts. The "3 sixes" cell has $E_4 = 0.46$, which breaks our rule that every cell needs $E_i > 1$.
- The usual fix is to collapse sparse categories. Here we combine "2 sixes" and "3 sixes" into a single "2 or more" category.

```
p_2plus <- 1 - dbinom(0, 3, 1/6) - dbinom(1, 3, 1/6)
Expected2 <- c(dbinom(0:1, size = 3, prob = 1/6), p_2plus)
chisq.test(x = c(47, 35, 18), p = Expected2)
```

```
##
##  Chi-squared test for given probabilities
##
## data:  c(47, 35, 18)
## X-squared = 17.192, df = 2, p-value = 0.0001849
```

- Now every expected count is above 5, and the conclusion is unchanged.

Parting humor, courtesy of the Comedian Erik Tanouye



Source: The Escape Club, Will Smith